

Strategic Design at Scale

DDD Patterns for Integration-Heavy Domains



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AS/400 RPG IV ACCT MAINT

ACCOUNT NO.: 00418822
CUSTOMER: MARTINEZ, J
BALANCE: 12,540.00
STATUS: ACTIVE

F3=Exit F5=Refresh F12=Cancel

RPG



Java

Java

Java

“That book made me look good.”

ERP. CRM. Forecast.

built in-house. all three.

Everyone knew how it worked.
Explaining it was harder.



CONTENT MANAGEMENT?



THE RULES EXIST

Accounting

Call routing

Manufacturing

Shipping



THE RULES EXIST

Accounting

Call routing

Manufacturing

Shipping

THE RULES ARE NEGOTIABLE

Logo position?

Popup?

Personalize?

Experiment?

Which audience?

Which message?

The innocent website

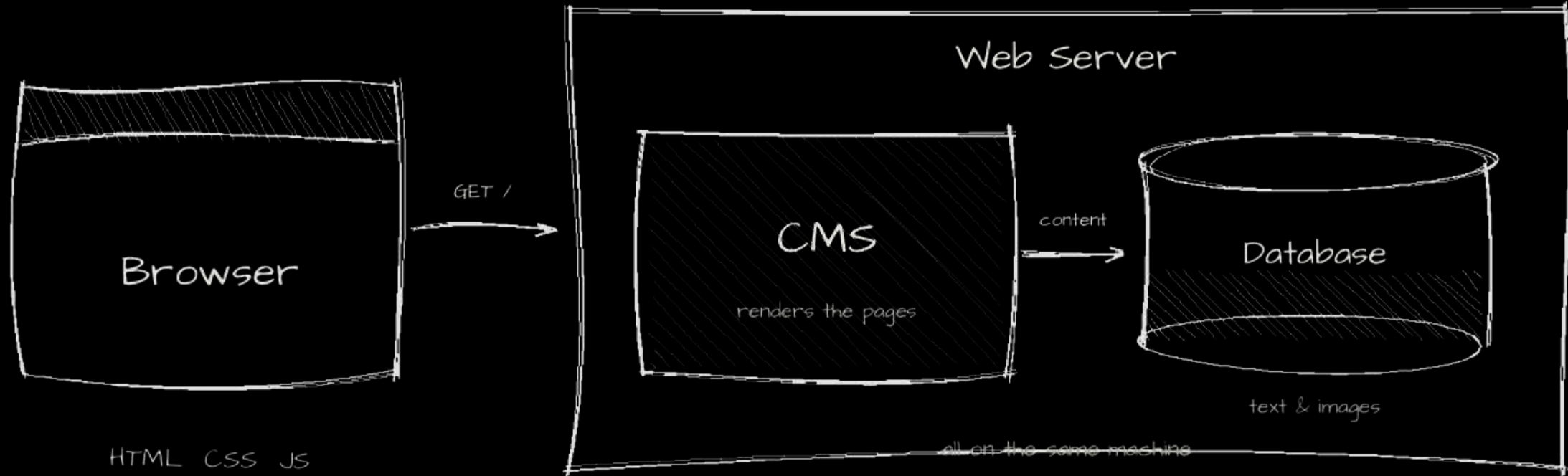


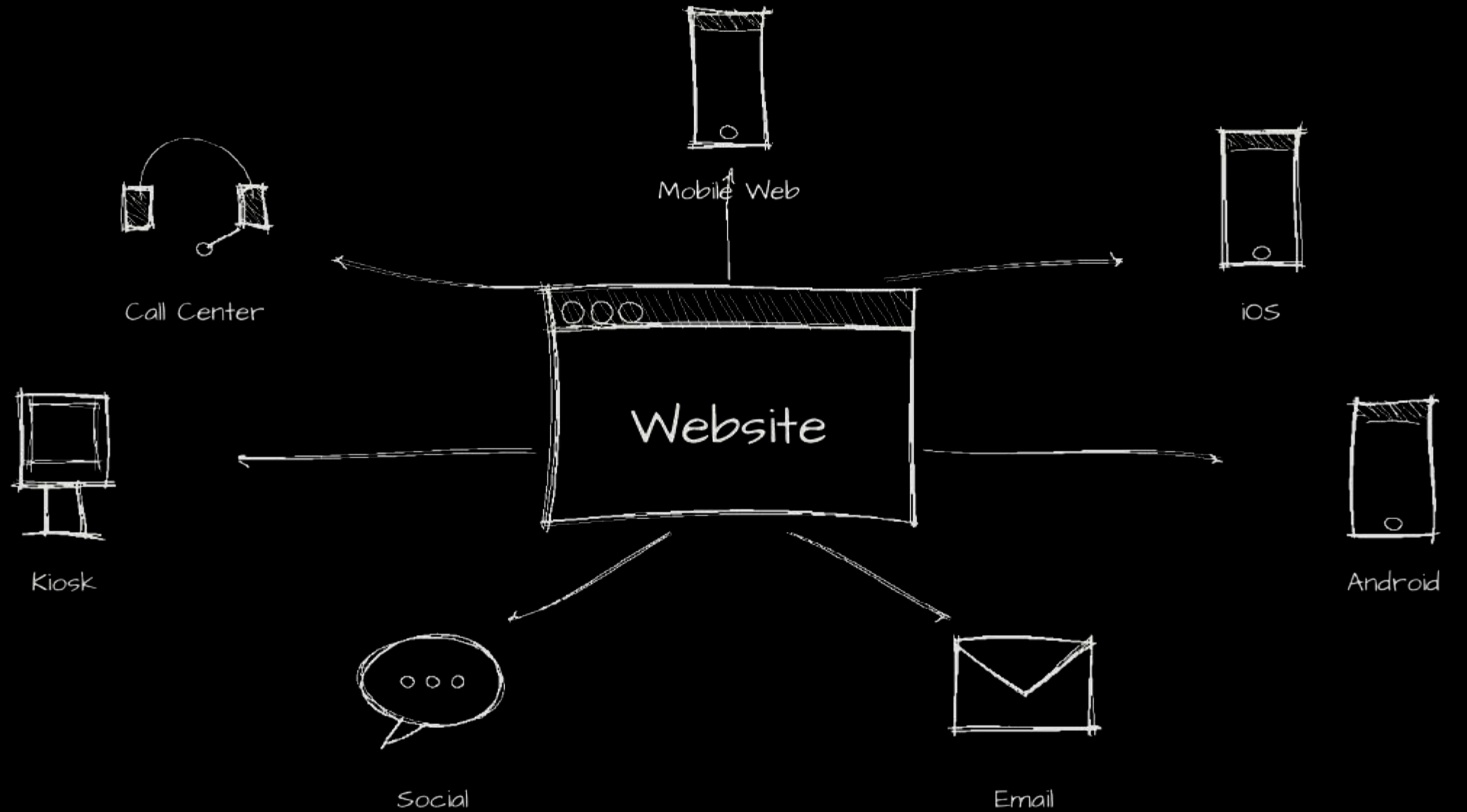
fig 1

One experience becomes many



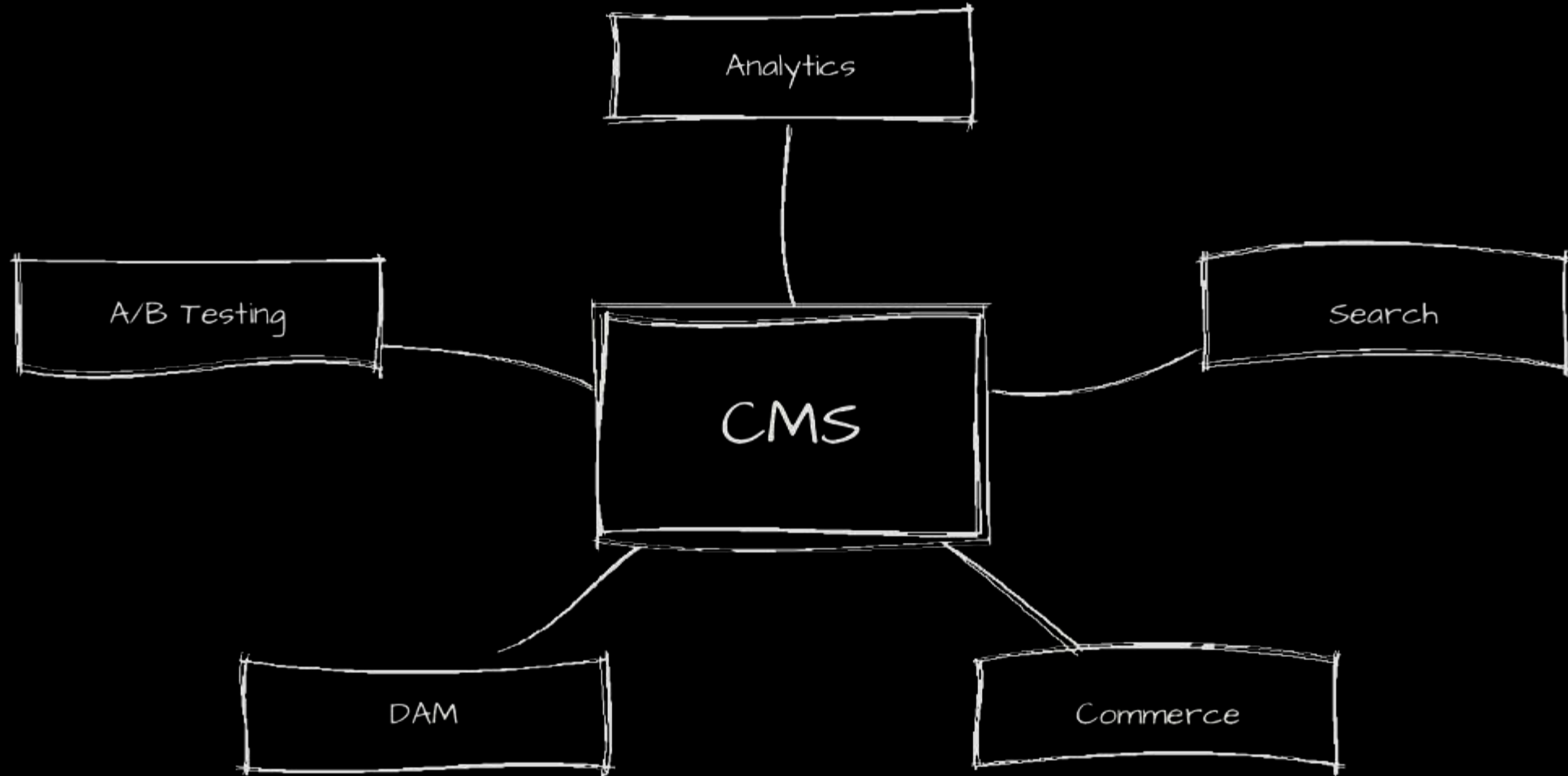
Website

One experience becomes many

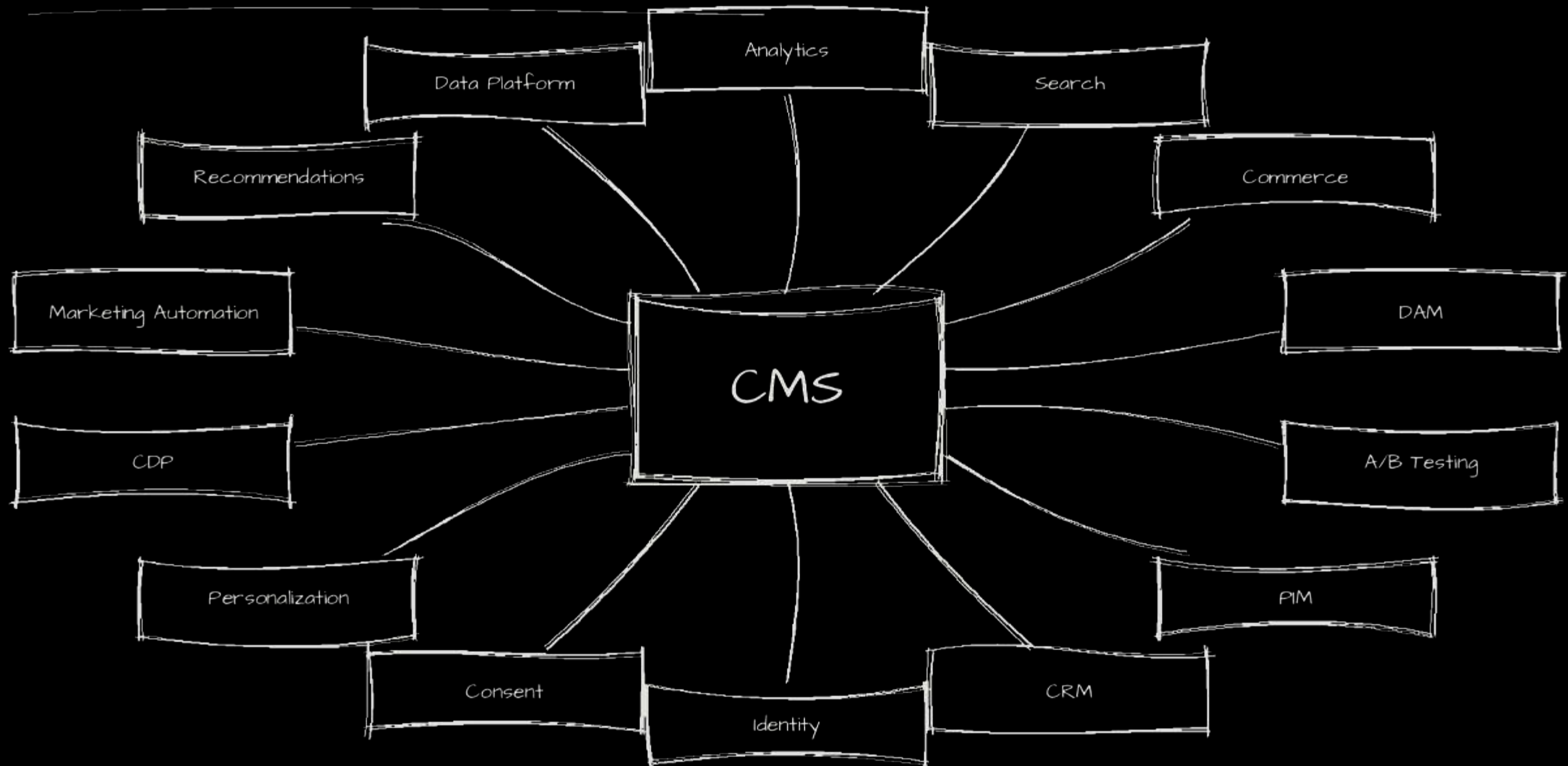


same content, many front doors

Every problem got a platform

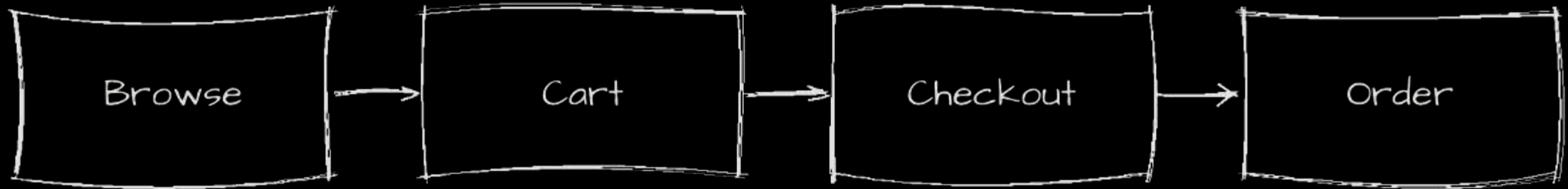


Every problem got a platform

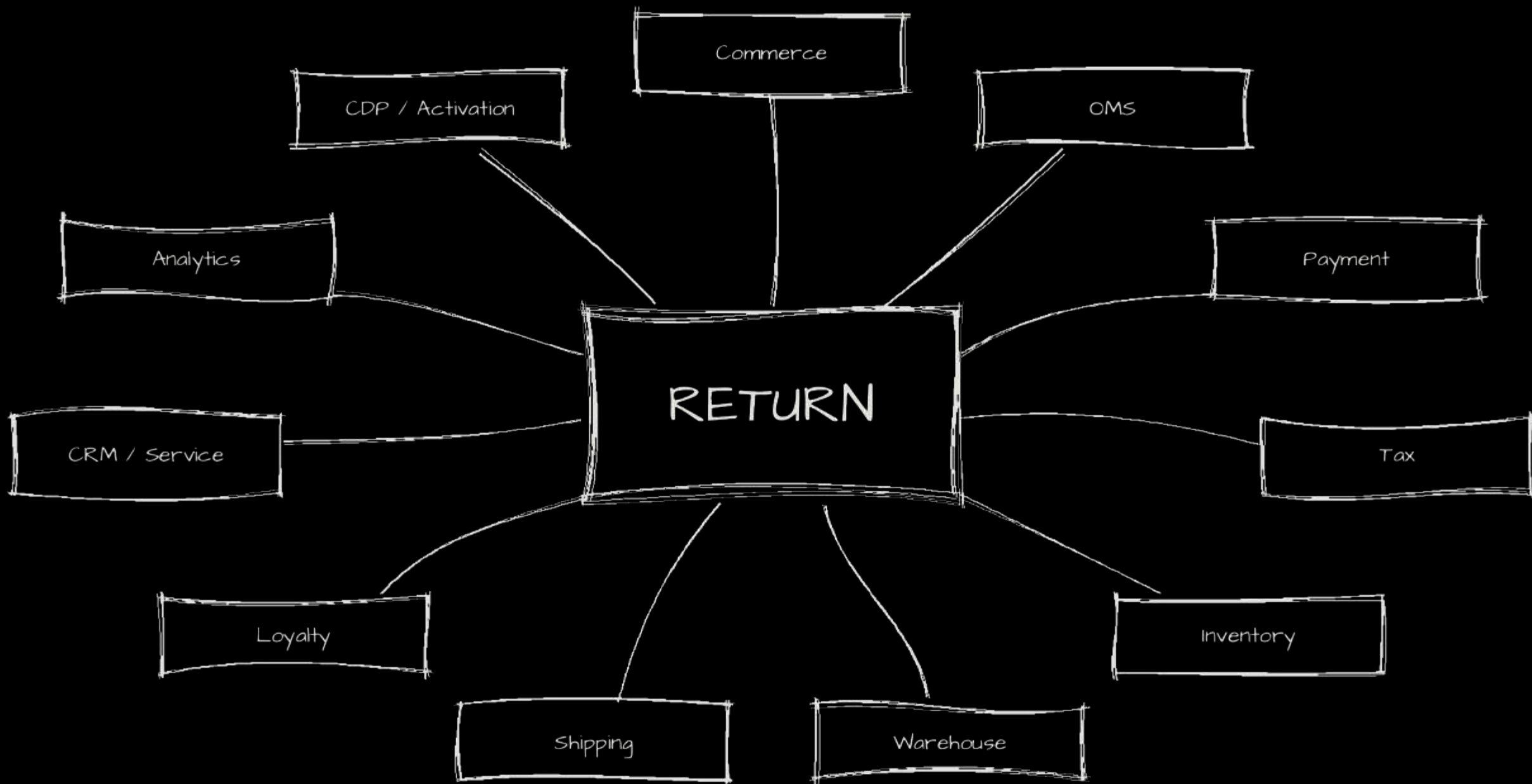


BUY

BUY



RETURN



The ecosystem



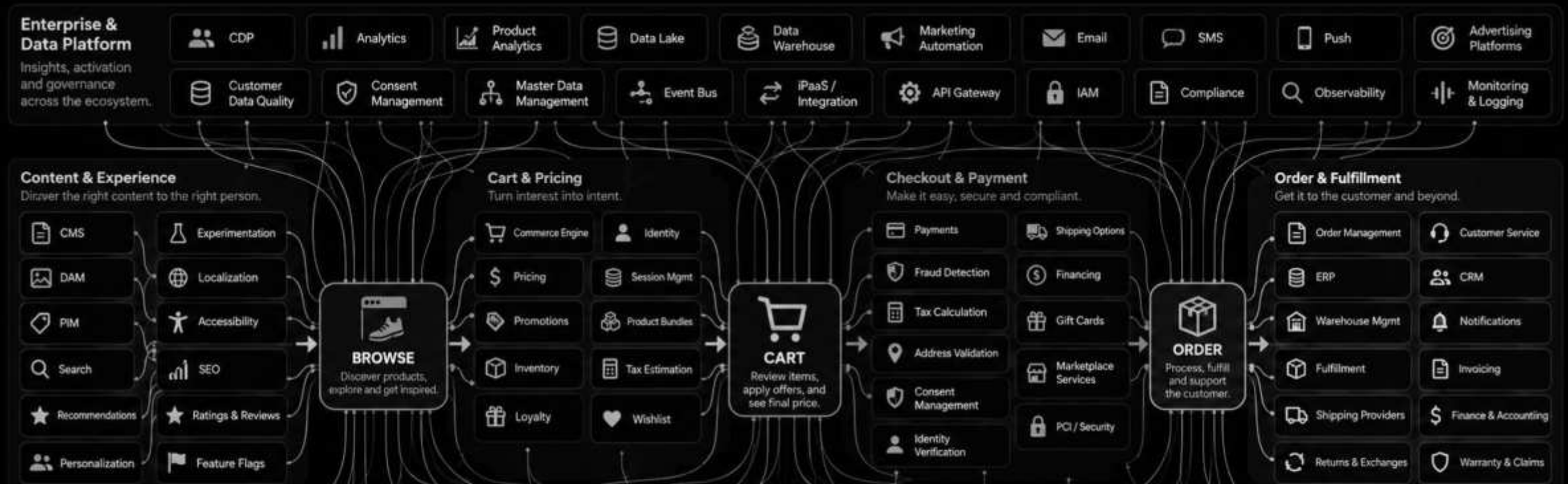
The ecosystem



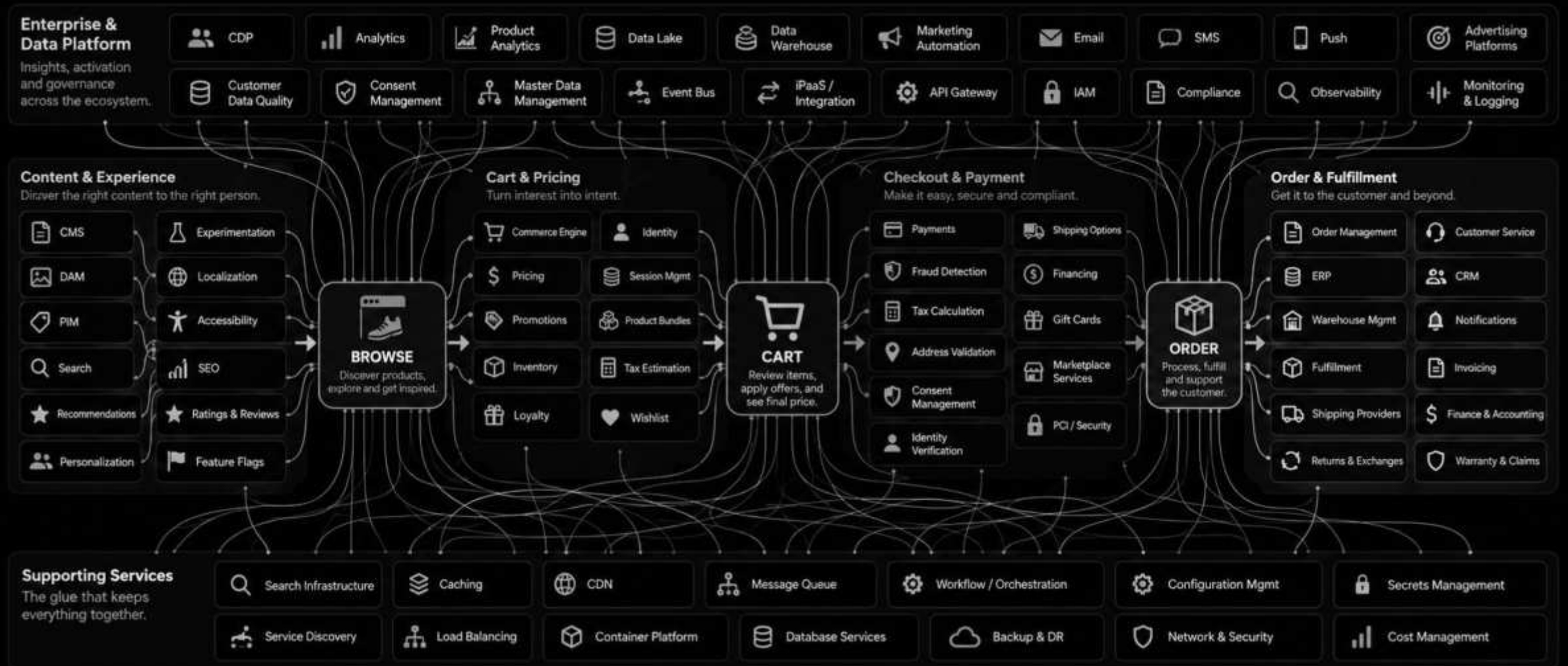
The ecosystem



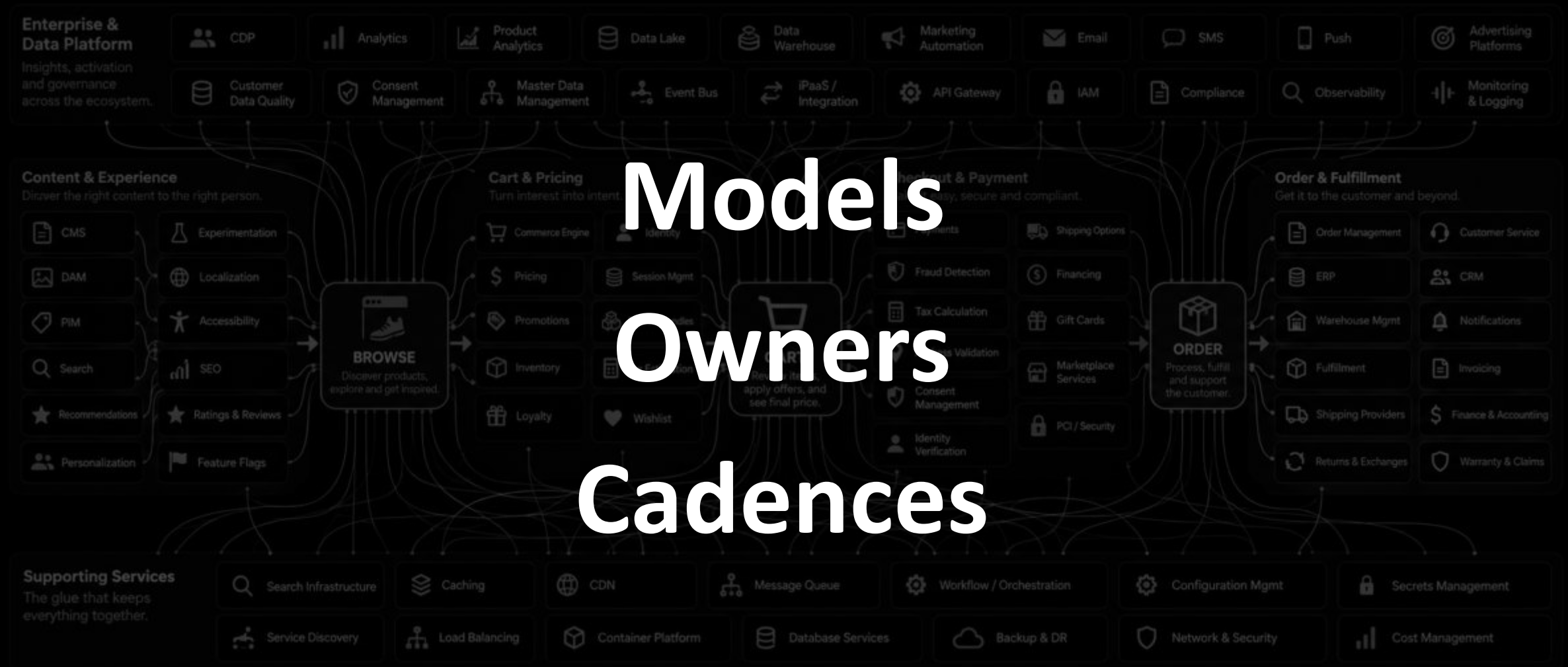
The ecosystem

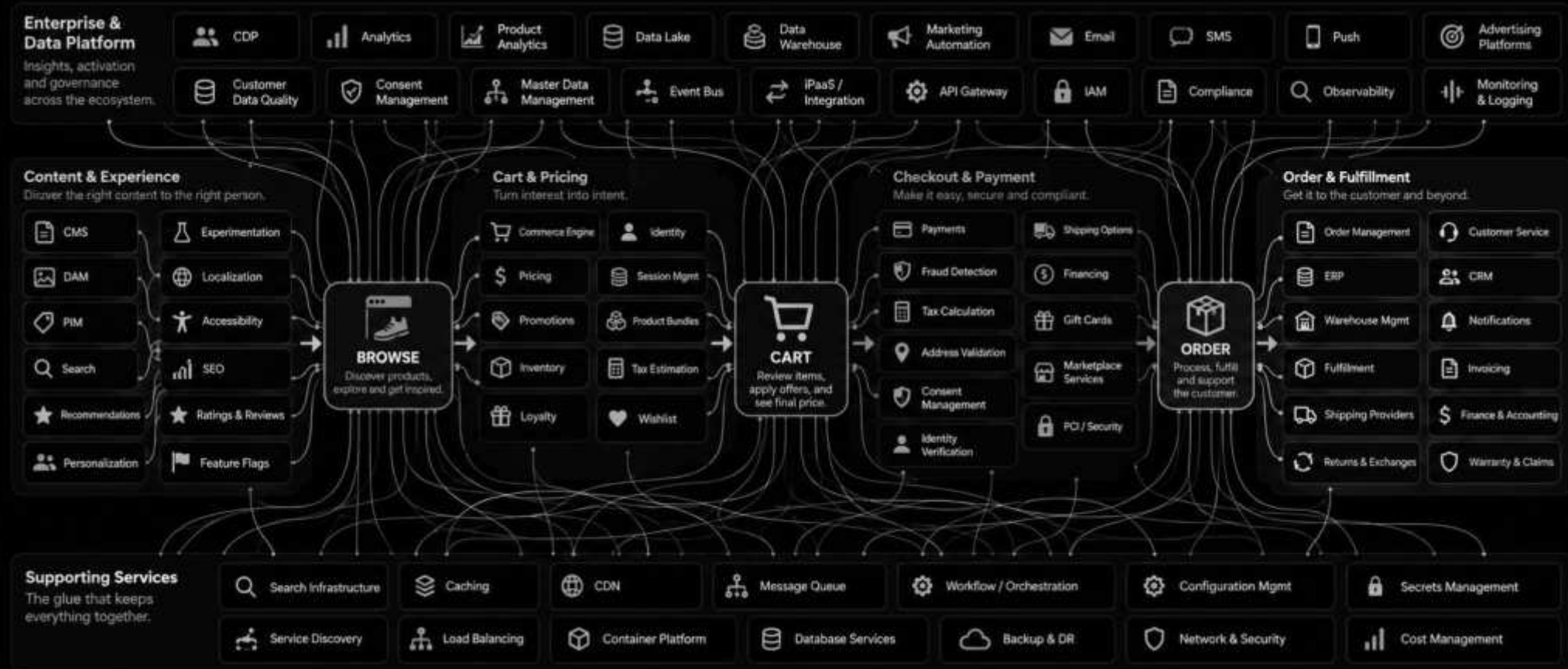


The ecosystem



The ecosystem





All of this... to buy—and return—a pair of shoes.

A black and white photograph of a person in a trench coat standing in a narrow, dark alleyway between brick buildings. The person is seen from behind, looking down a path that leads to a door at the end. A large shadow of a question mark is cast on the right wall. The text "COMPLICATED or COMPLEX?" is overlaid in the center.

COMPLICATED
or
COMPLEX?



$$2 + 2 = 4$$



Rubik's Cube



100-city Traveling Salesperson Problem

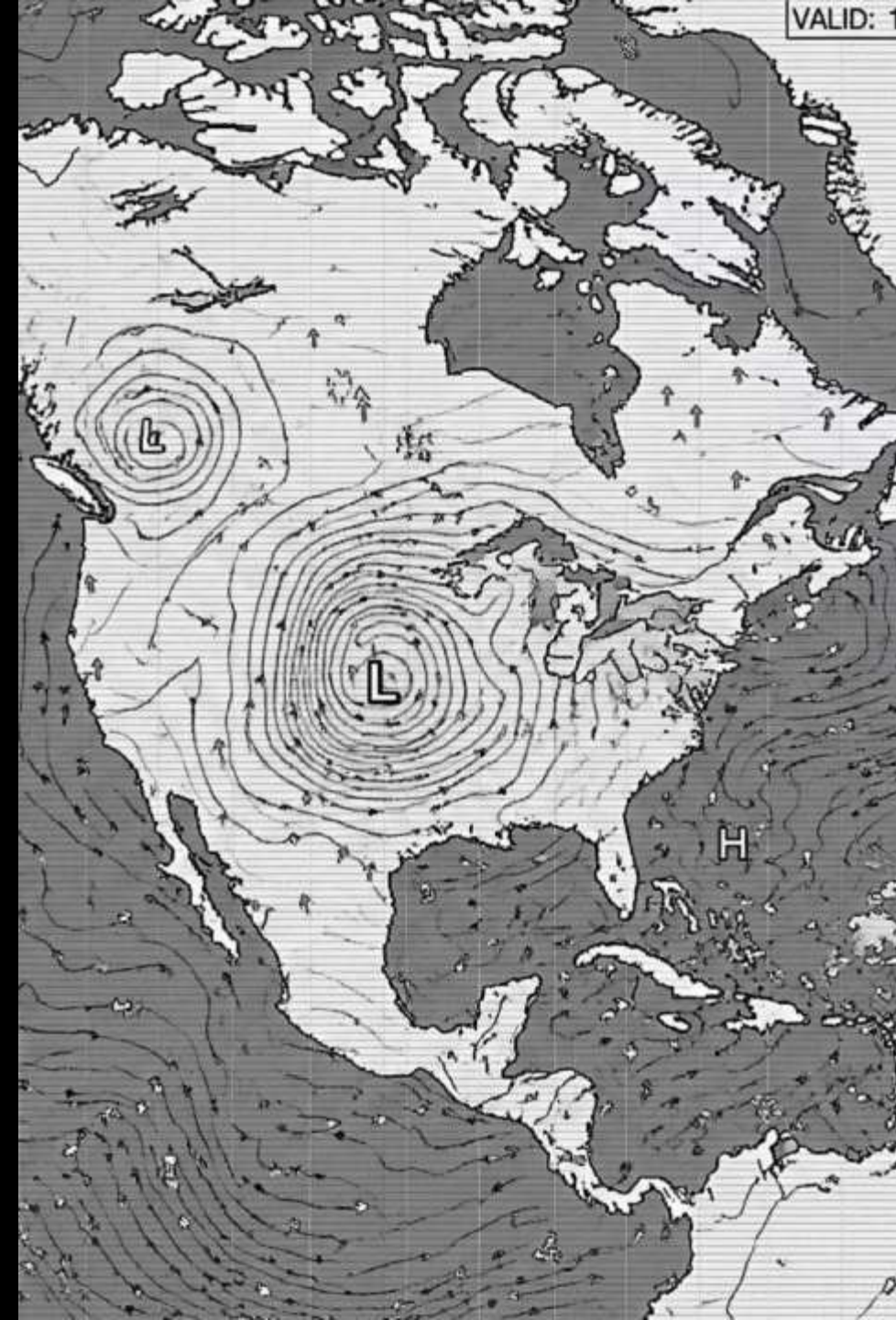
Known rules.

A solution exists.

Potentially enormous computation.

Temperature
Pressure
Ocean
Humidity
Terrain
Time

The parts interact.
The outcome emerges.



Simple (Sencillo) — Complicated
Simple — COMPLEX



That humble website
turned out to be
COMPLICATED and COMPLEX.



Essential COMPLEXITY

Customer

Product

Price

Inventory

Order

Consent

Return



Acquisition 2017

Accidental COMPLEXITY

Legacy CRM

Excel

New Middleware

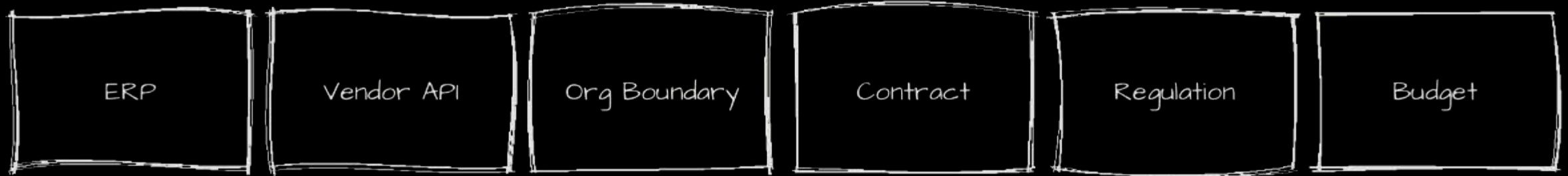
Old Middleware

Migration

Regional Platform

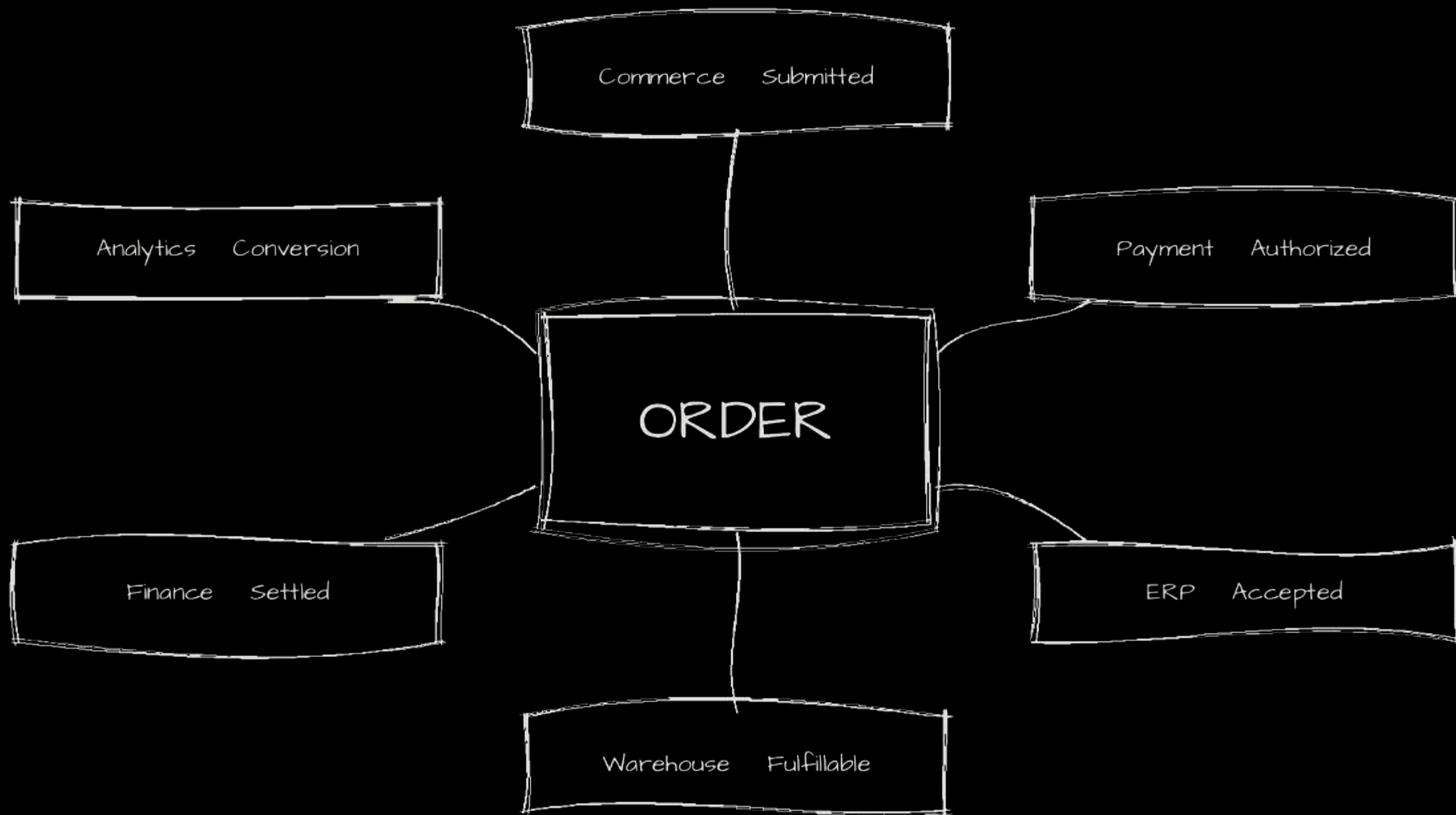
FINAL_product_master_v7_REALFINAL.xlsx

ACCIDENTAL COMPLEXITY + TIME = CONSTRAINT



HOW DO I MODEL THE DOMAIN?

**HOW DO I MODEL A DOMAIN
THAT HAS ALREADY BEEN MODELED
FIFTEEN DIFFERENT TIMES?**



Ubiquitous... where?

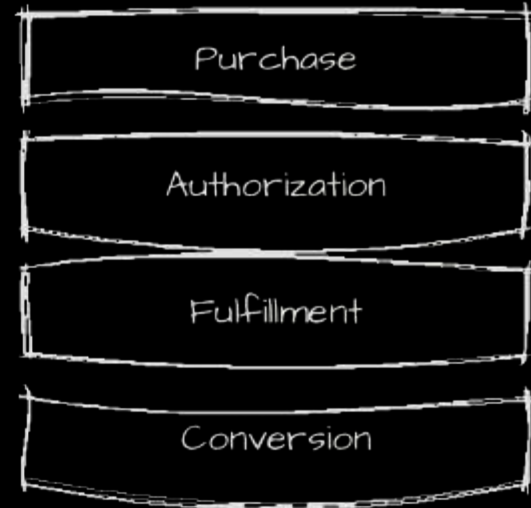
CUSTOMER



PRODUCT



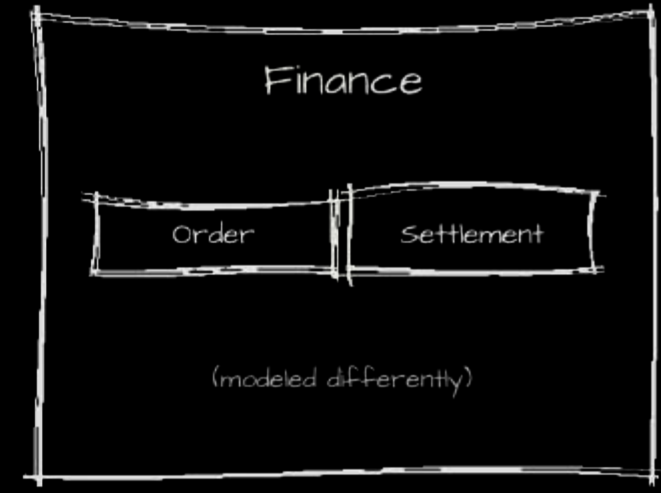
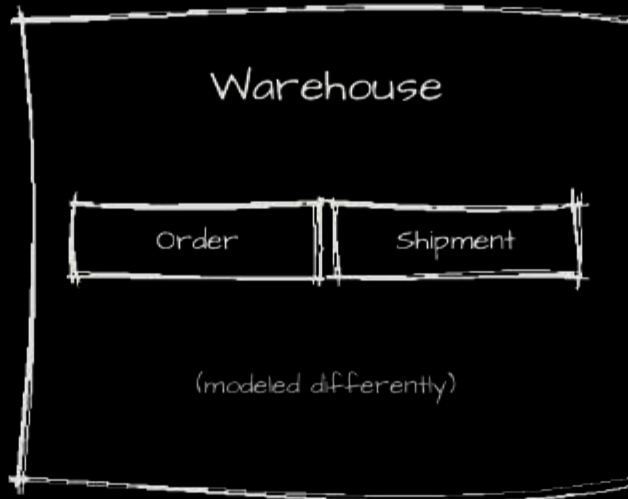
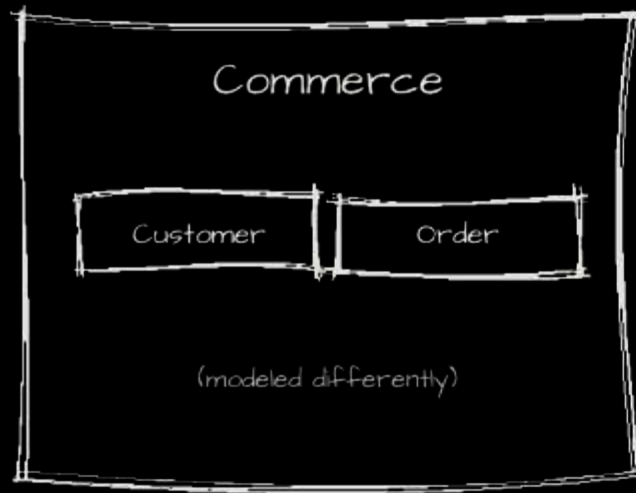
ORDER



Language is ubiquitous inside a context.

Organization

Vendor



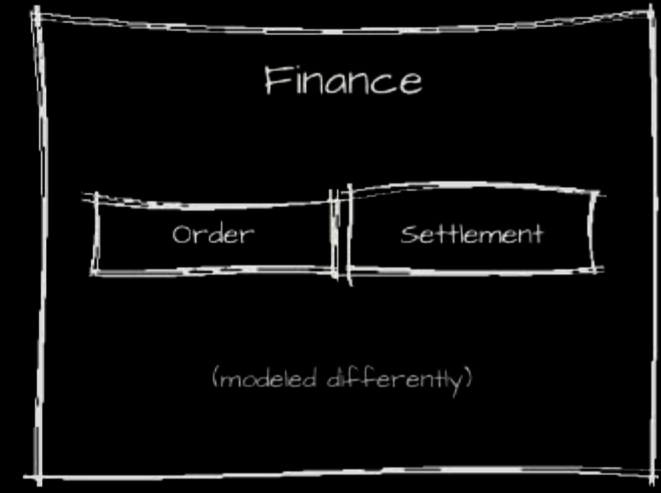
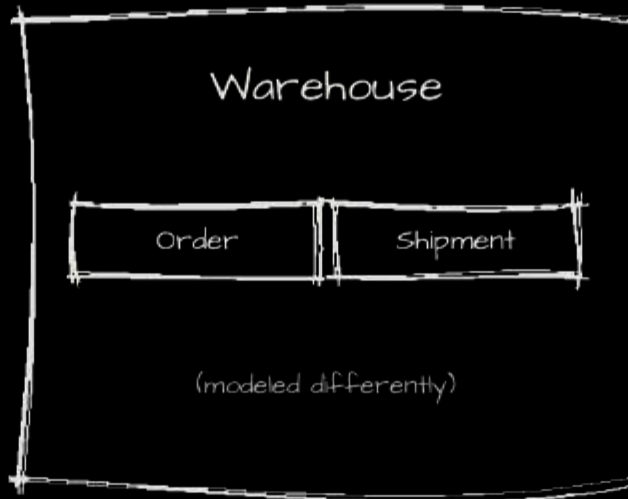
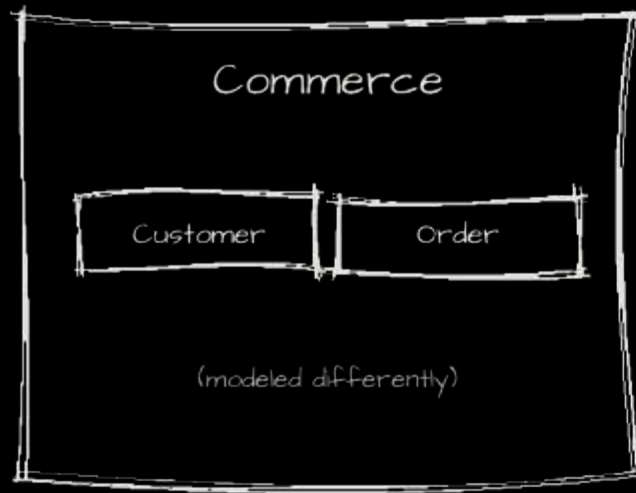
Regulation

Geography

YOU MAY NOT CONTROL THE BOUNDARIES.

Organization

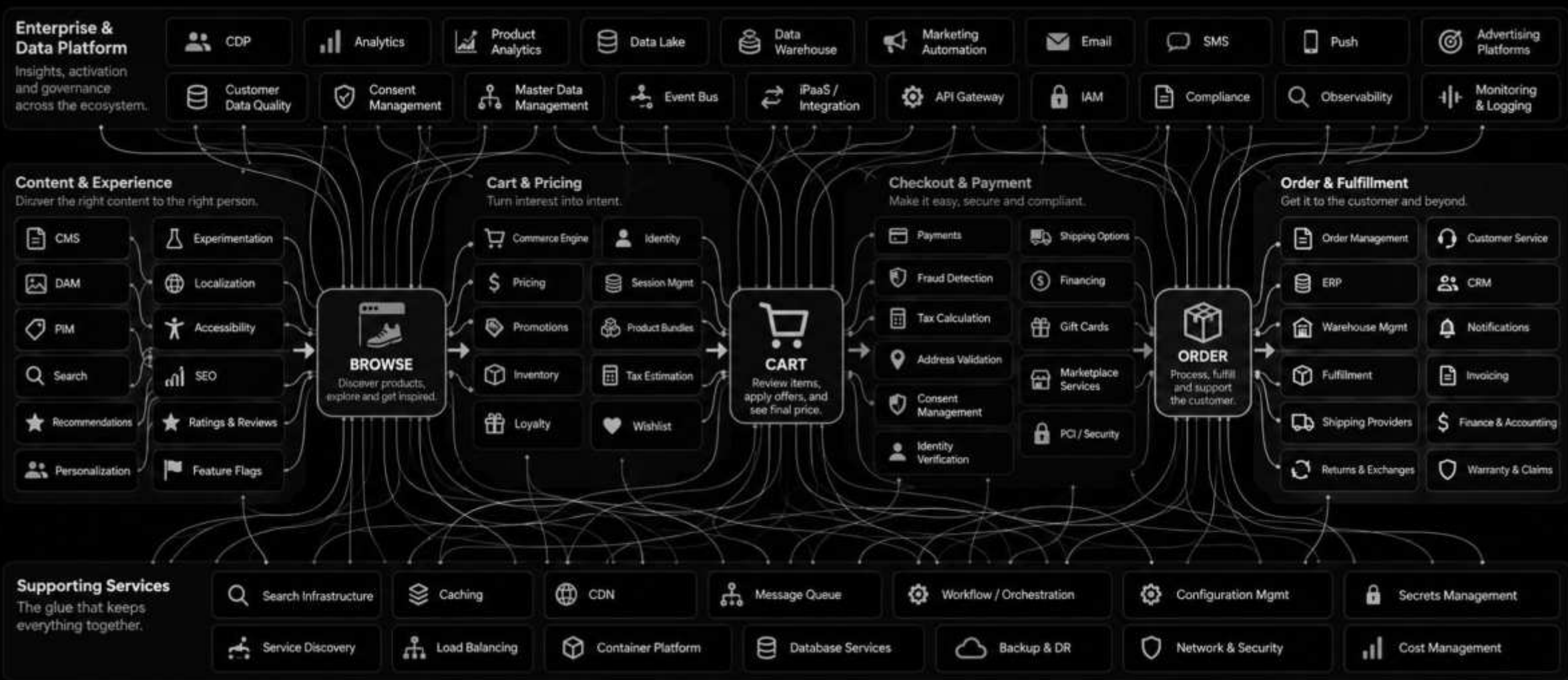
Vendor

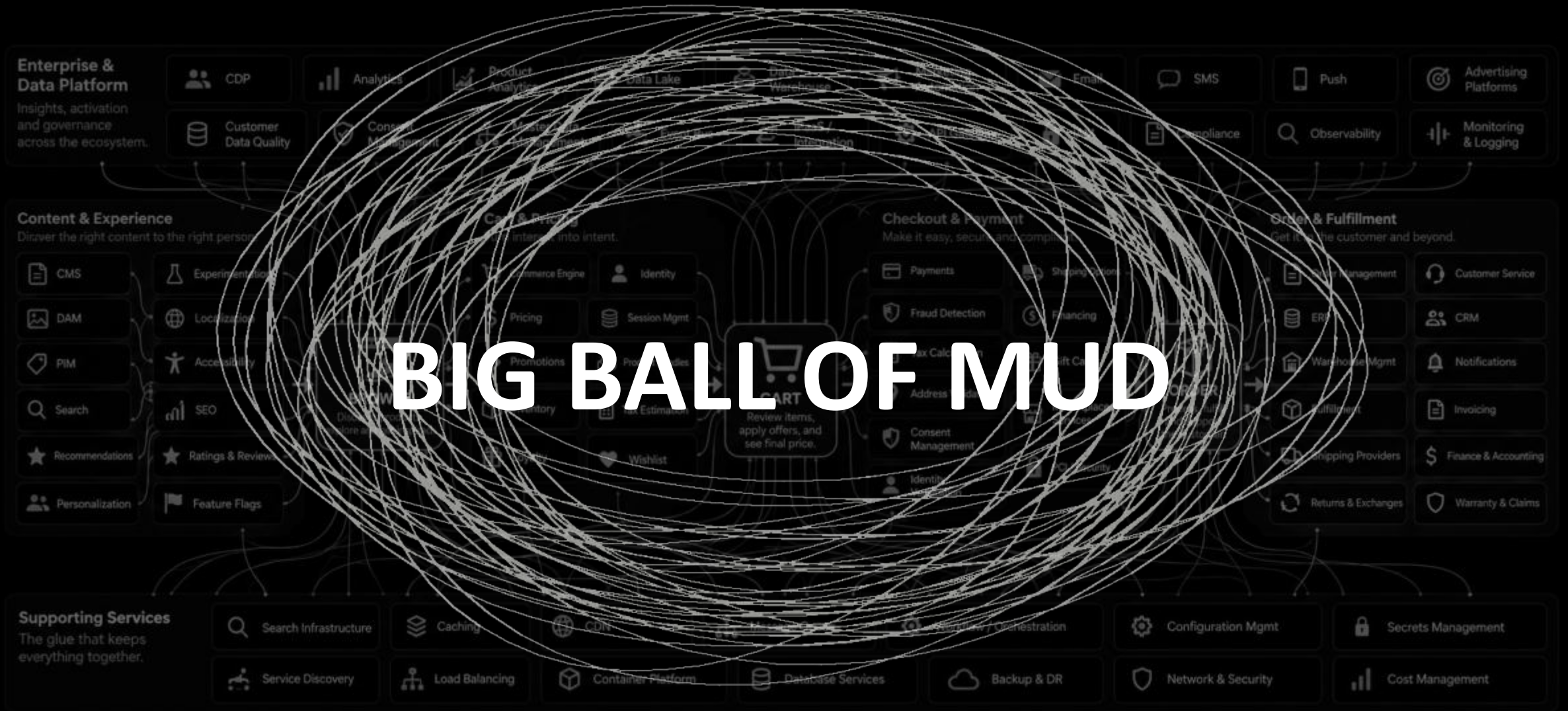


Regulation

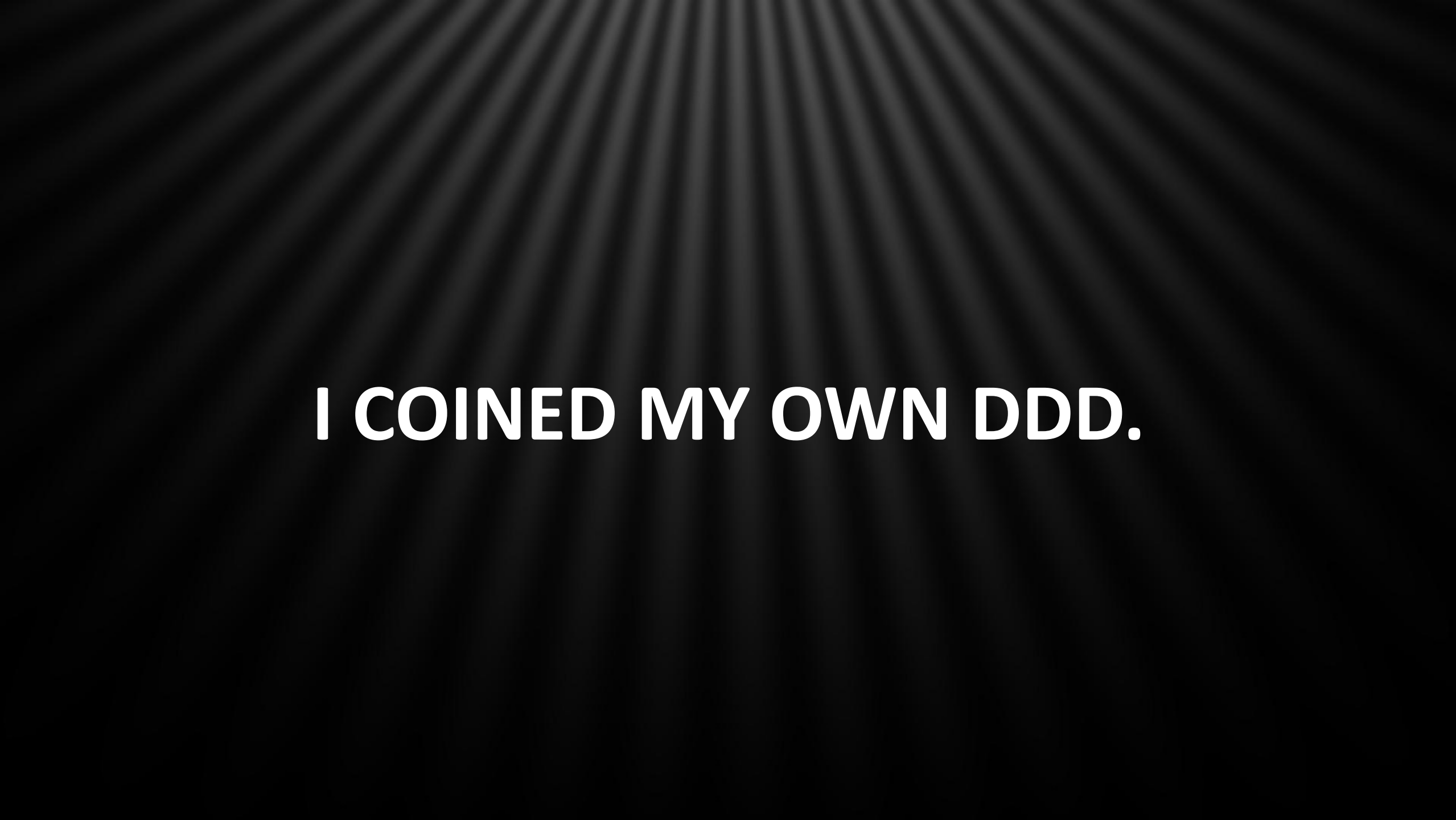
Geography

YOU STILL NEED TO MAKE THEM EXPLICIT.





BIG BALL OF MUD



I COINED MY OWN DDD.



I COINED MY OWN DDD.

Well... PDD.

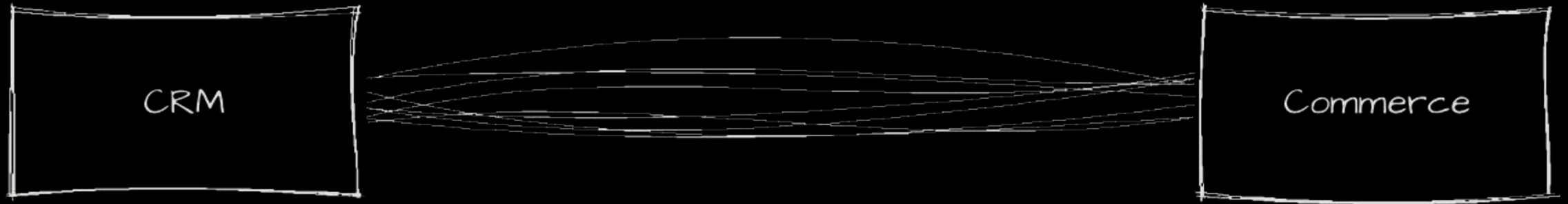
PROCUREMENT-DRIVEN DESIGN

	vendor A	vendor B	vendor C	vendor D	vendor E
price	7	6	9	5	8
support terms	6	8	8	7	6
security review	8	7	9	6	7
integrates with what we own	5	6	9	4	5
local partner	4	7	8	6	9
enterprise agreement	3	5	9	2	4
total	33	39	52	30	39

no row for: what this does to our model

THE RFP PRICED THE BOXES.

nobody priced the arrows.



- field mapping
- error handling
- monitoring
- the one person who understands both sides
- identity reconciliation
- replay at 2am
- the upgrade that breaks it

every arrow is a translation a translation is domain logic.

A black and white photograph showing two people shaking hands over a table. On the table, there is a laptop, a glass of coffee, and some papers. The scene is dimly lit, creating a moody atmosphere. The text "I FOUGHT THIS FOR YEARS." is overlaid in white, bold, sans-serif font across the center of the image.

I FOUGHT THIS FOR YEARS.

A black and white photograph of a man sitting at a desk, looking down at his work. He is holding a small cup. The desk is cluttered with papers, books, and a lamp. The background is dark and out of focus.

WE WERE NOT BUILDING THE BEST SYSTEM.

We were building the one that made sense to that enterprise, that day.

and that is not a failure.

What Conway said



“...the very act of organizing a design team means that certain design decisions **have been already been made**, explicitly or otherwise.”



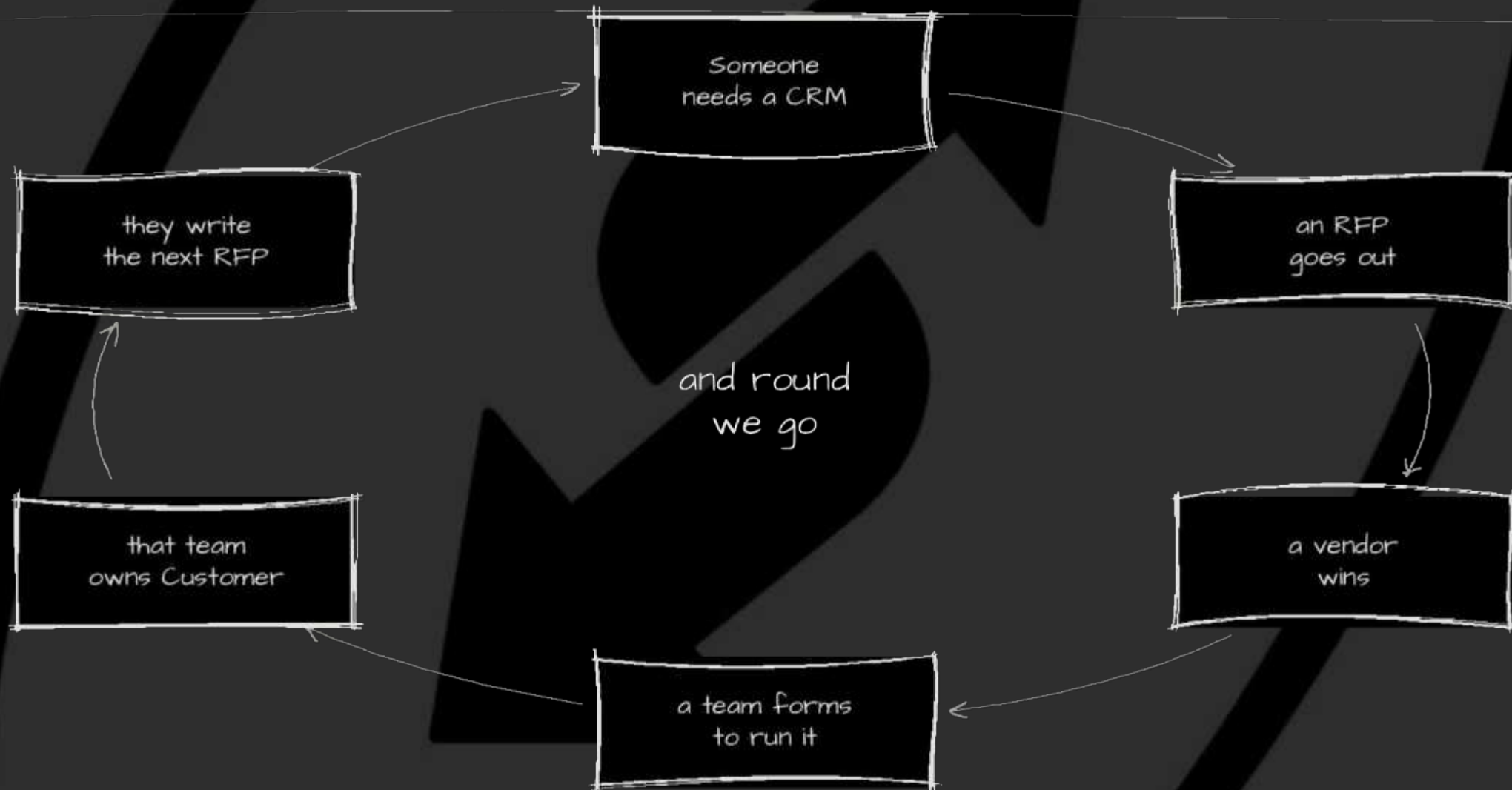
“Many of these choices may be more than design decisions; they may also be **personal decisions the designer makes about his own future.**”

Melvin Conway, 1968

ORGANIZATION PRECEDES ARCHITECTURE.

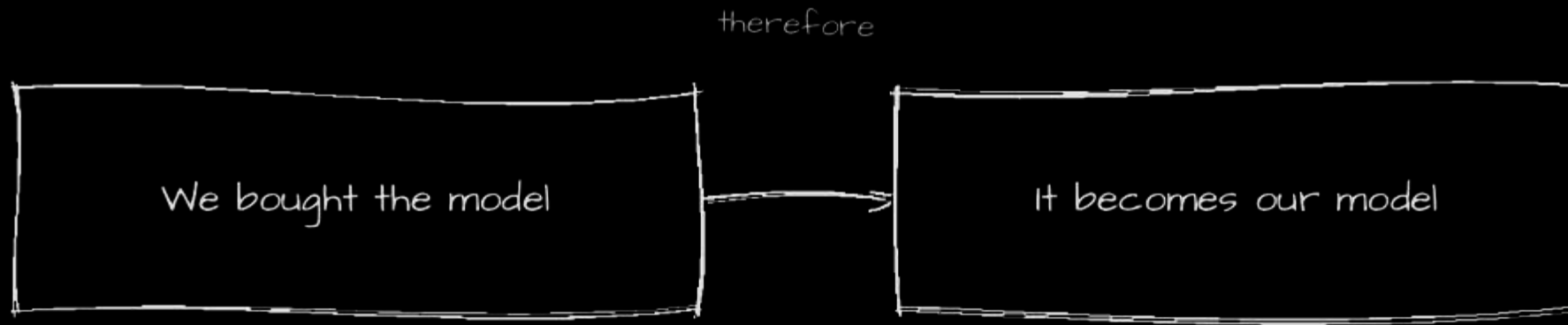
“...the availability of a design organization in need of work.”

Melvin Conway, 1968



ANTI-PATTERN 1

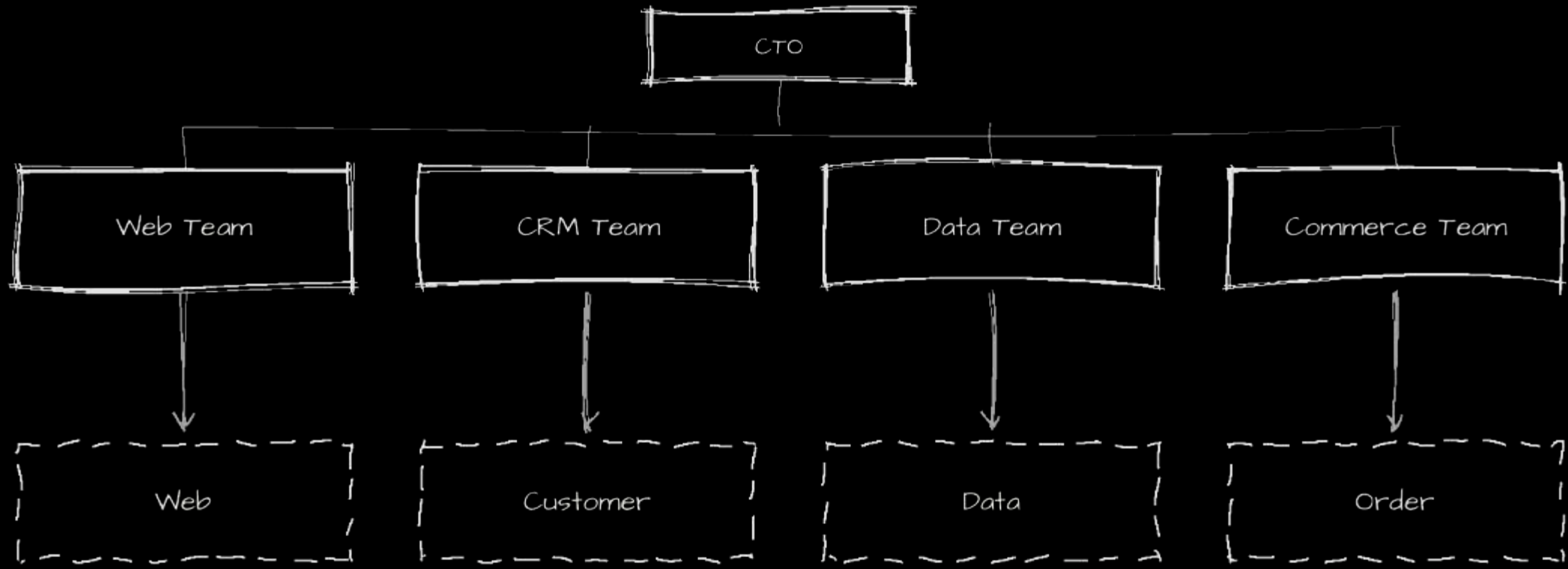
PROCUREMENT-DRIVEN DOMAIN



in DDD terms. Conformist - but nobody chose it

ANTI-PATTERN 2

THE ORG CHART BECOMES THE DOMAIN MAP

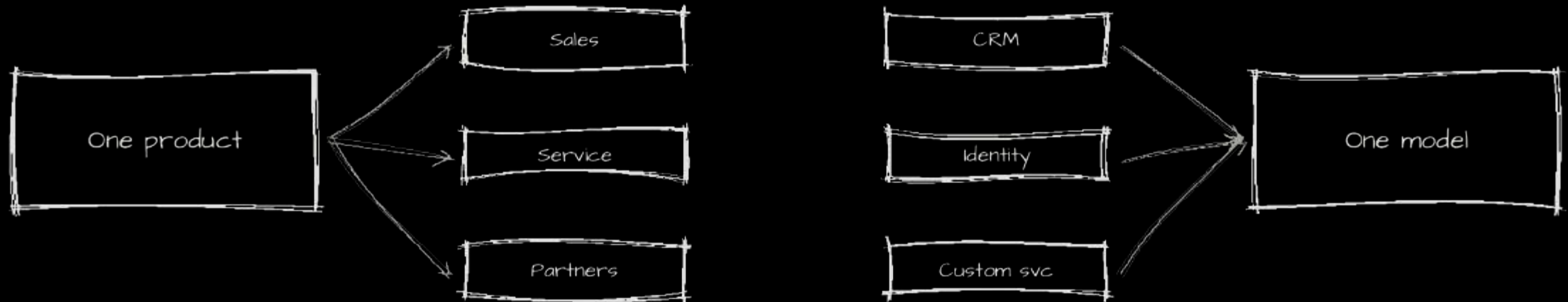


a context map is not an org chart

ANTI-PATTERN 3

PRODUCT = BOUNDED CONTEXT

~~Salesforce = CRM Context~~

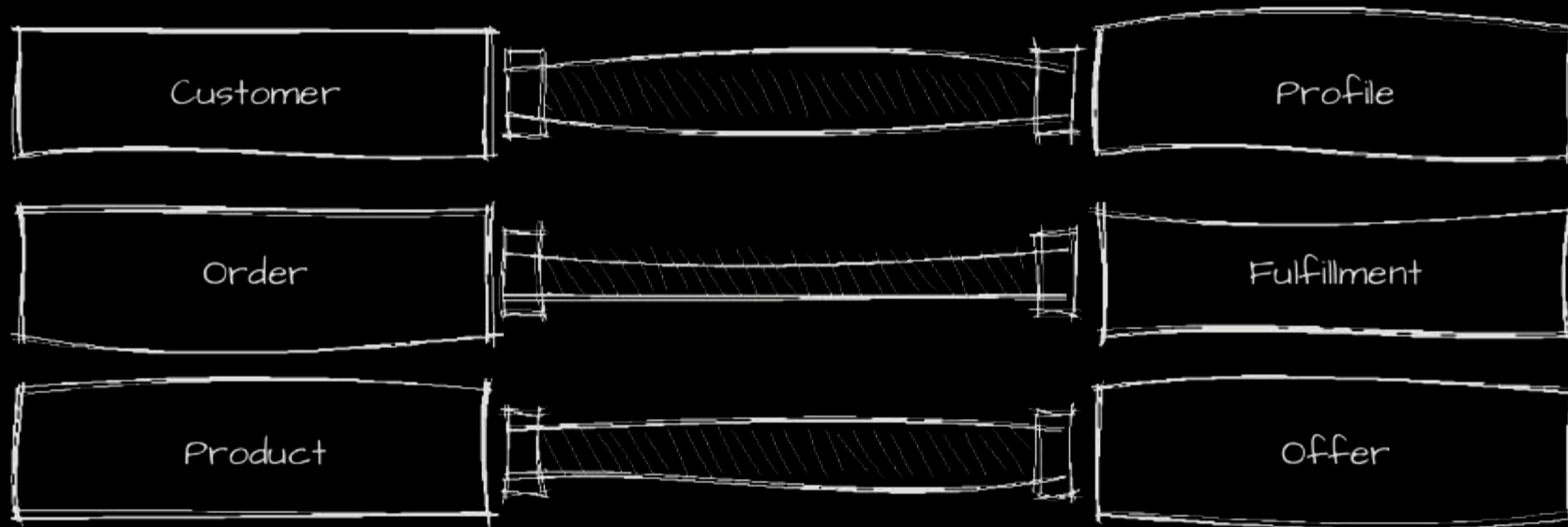


problem space: subdomain · solution space: bounded context · a product is neither

ANTI-PATTERN 4

“INTEGRATION IS PLUMBING”

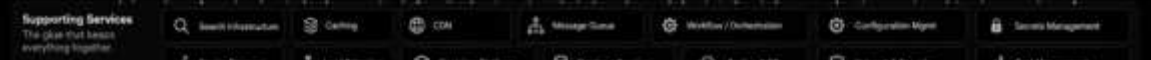
this seam already has names: ACL · Published Language · Open Host Service



Plumbing?

ANTI-PATTERN 5

ARCHITECTURE WALLPAPER

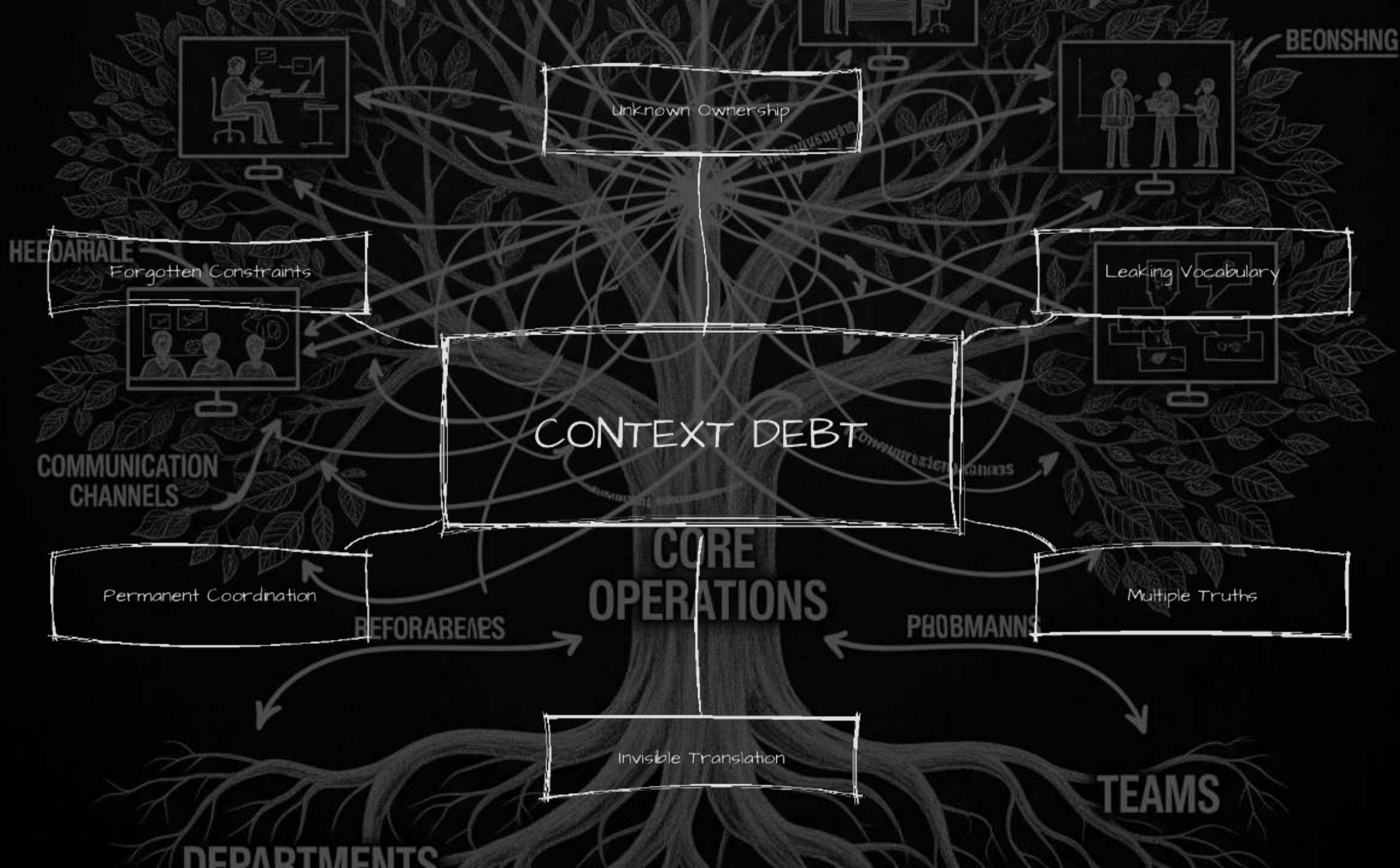


ANTI-PATTERN 6

THE PLATFORM WILL SOLVE IT

you bought the aircraft. who is flying it?

in DDD terms a context nobody can operate is a context nobody owns



SO... WHAT IS MY JOB?

THE MODELER IS A STRATEGIST



Why “strategist”?

ARCHITECTURE WORK

Systems

Components

Technology

Interfaces

STRATEGIC MODELING

Meaning

Boundaries

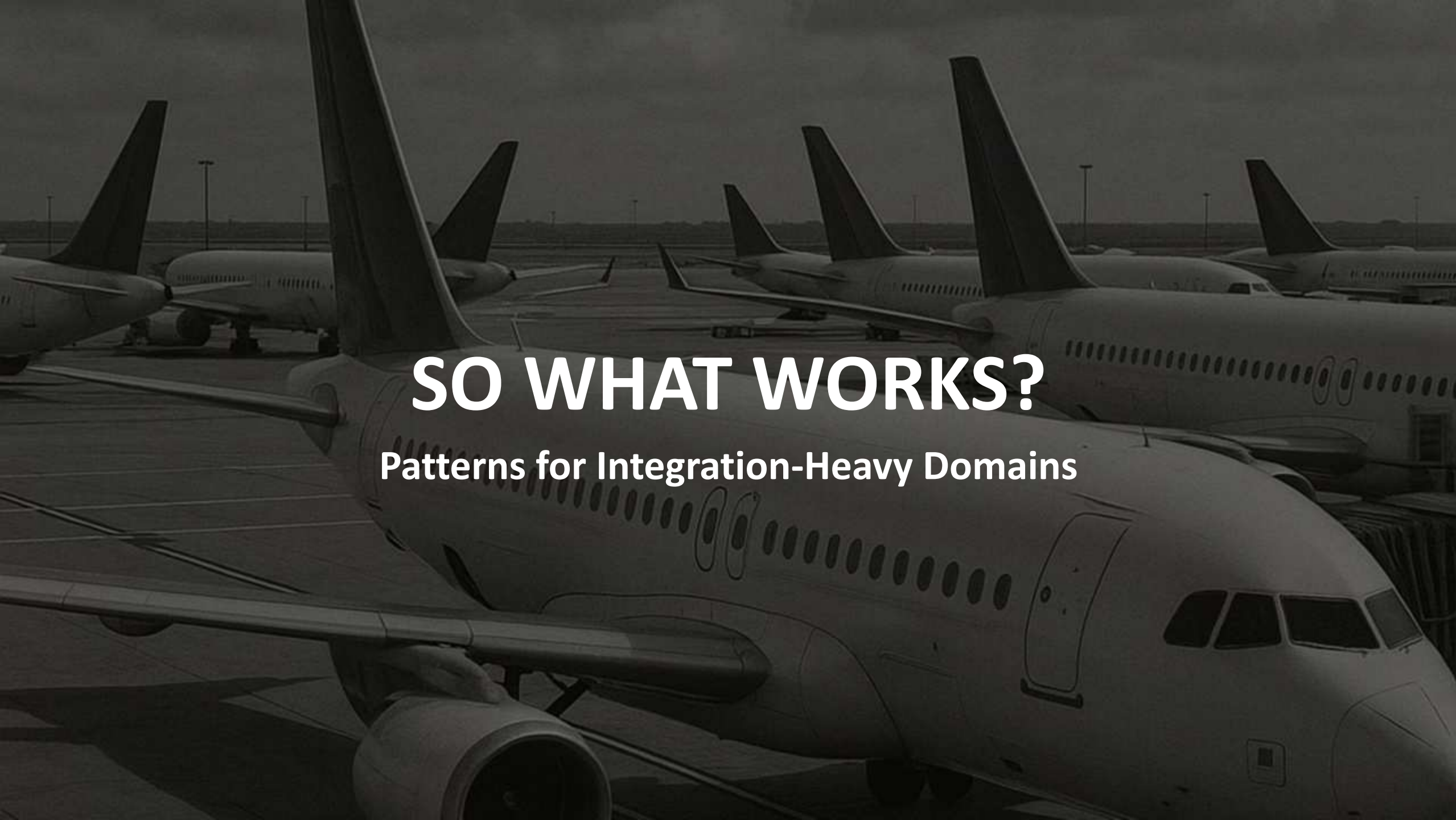
Relationships

Constraints

Choices

Evolution

Both matter. This talk is about the second.



SO WHAT WORKS?

Patterns for Integration-Heavy Domains

PATTERN 1

CAPABILITY FIRST. PRODUCT SECOND.

pick the tool after you know the job.

what is this capability,

and is it core?



PATTERN 1 · CAPABILITY FIRST. PRODUCT SECOND.

CORE, SUPPORTING, GENERIC

CORE	build it - usually on top of a platform own the model, rent the plumbing.
SUPPORTING	buy it, then shape it.
GENERIC	buy the most boring thing. do not customise it.

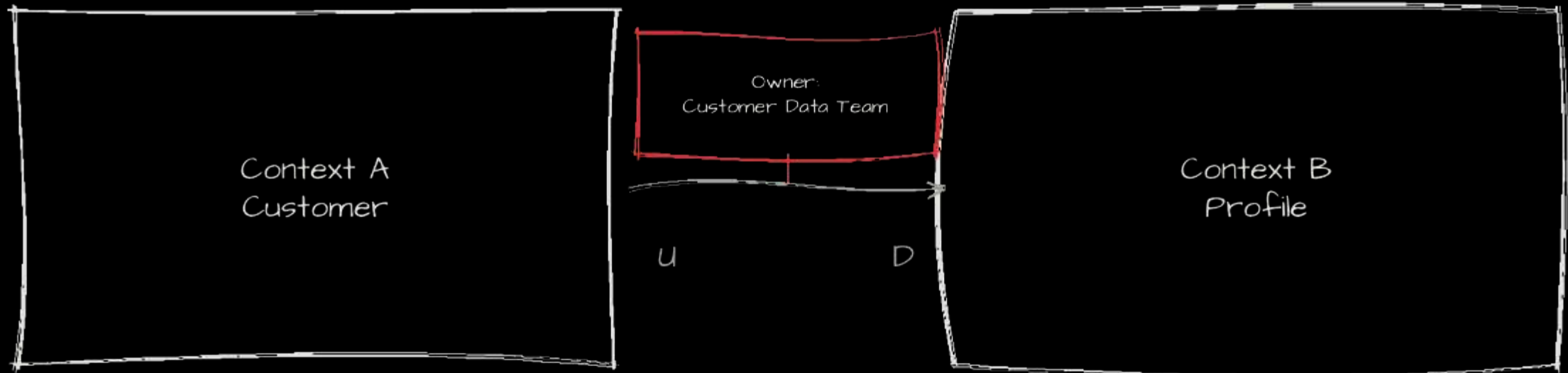
before any of it: what happens if we don't have this at all?

'generic' on the vendor's website is not generic inside your company

and who operates it is a different question from who owns the model

PATTERN 2

OWN THE BOUNDARY



we are good at drawing U and D. we never write who

PATTERN 2 · OWN THE BOUNDARY

FOUR THINGS YOU CAN DO, AND WHAT EACH COSTS

take their word for it

you pay in meaning, their words quietly become yours.

translate at the edge

you pay for a team forever. the only one you can do alone.

agree a contract

you pay in time, and only if you have something they want.

don't connect them at all

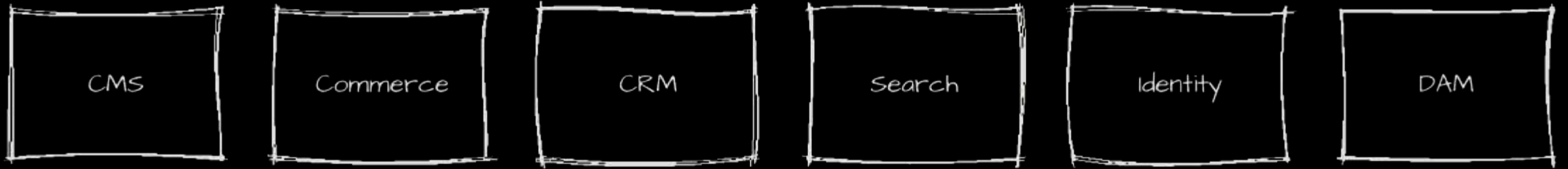
you pay in consistency, on purpose. often the cheapest.

the books call these conformist, anticorruption layer, customer-supplier, separate ways.

the name matters less than writing down which one you picked.

PATTERN 3

NOT ALL SQUARES ARE THE SAME SIZE

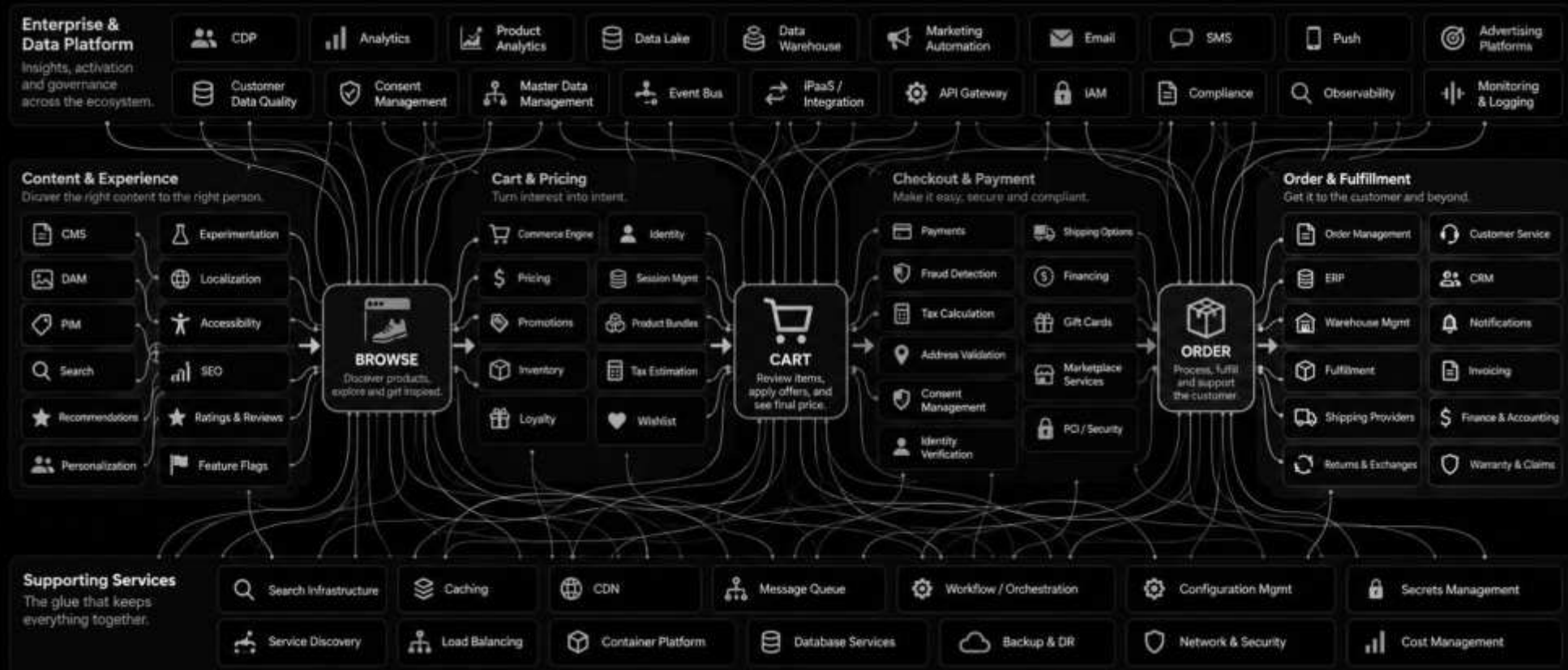


drawn the same. not the same.

wallpaper hides what matters. the invoice shows it.

PATTERN 3 · NOT ALL SQUARES ARE THE SAME SIZE

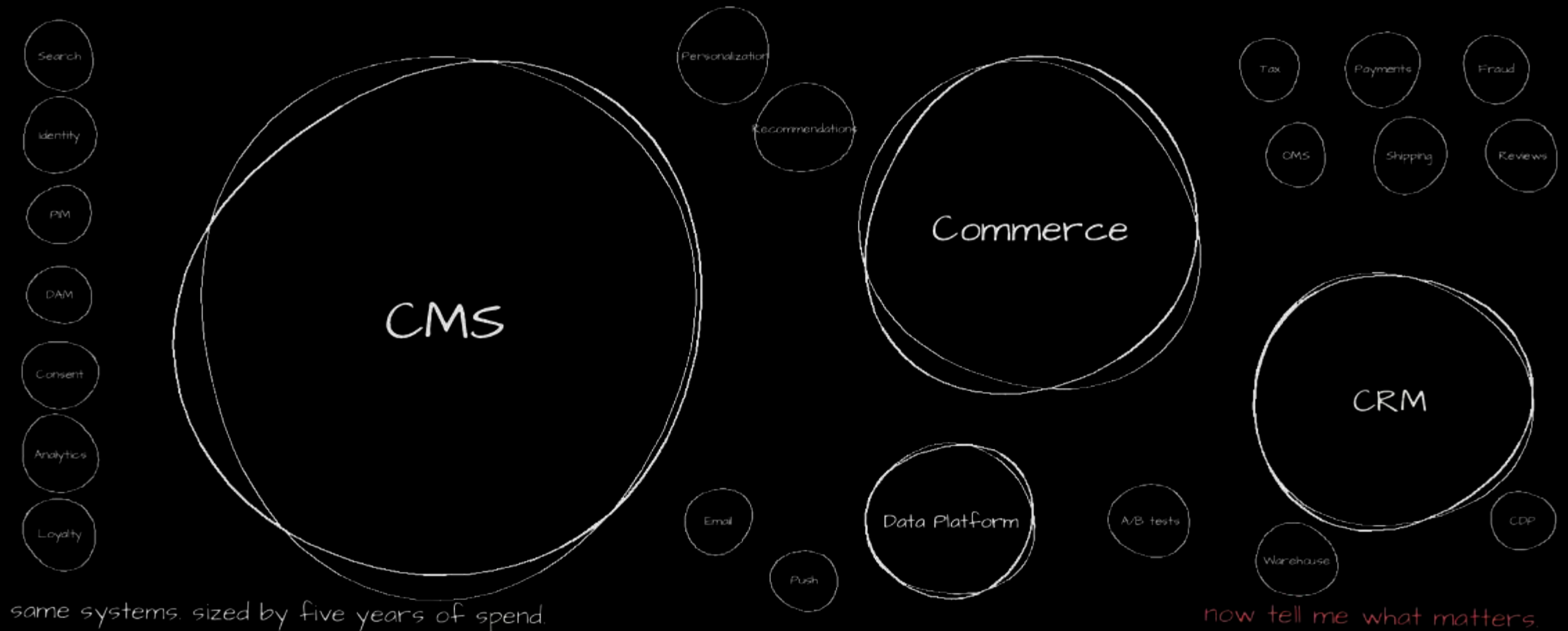
FIFTY BOXES. ONE SIZE.



fifty boxes. one size. which three matter?

PATTERN 3 · NOT ALL SQUARES ARE THE SAME SIZE

THE SAME SYSTEMS, SIZED BY WHAT WE PAY



PATTERN 3 · NOT ALL SQUARES ARE THE SAME SIZE

AND THE TOLL AT EVERY CROSSING

Customer Profile - nobody owns the translation	14 days
two live definitions of Order	9 days
"Contact" means three things in three systems	6 days
a boundary nobody remembers drawing	4 days
this project	33 days

same invoice, next project, and the one after that

PATTERN 4

SDRs



ADR

Architecture Decision Record

SDR

Strategic Decision Record

only just thought of this.

WHAT AN SDR HOLDS

CONTEXT

What problem are we solving?

CONSTRAINTS

What couldn't we change?

STRATEGIC TENSION

What models / teams / constraints conflict?

CONSEQUENCES

What do we gain / accept?

DECISION

What did we choose?

REVISIT WHEN

What would invalidate this decision?

BOUNDARY IMPACT

What meaning or ownership changes?

**ONE KEEPS THE HOW. THE OTHER KEEPS THE
WHY.**

ADR

Why does this integration use
batch?

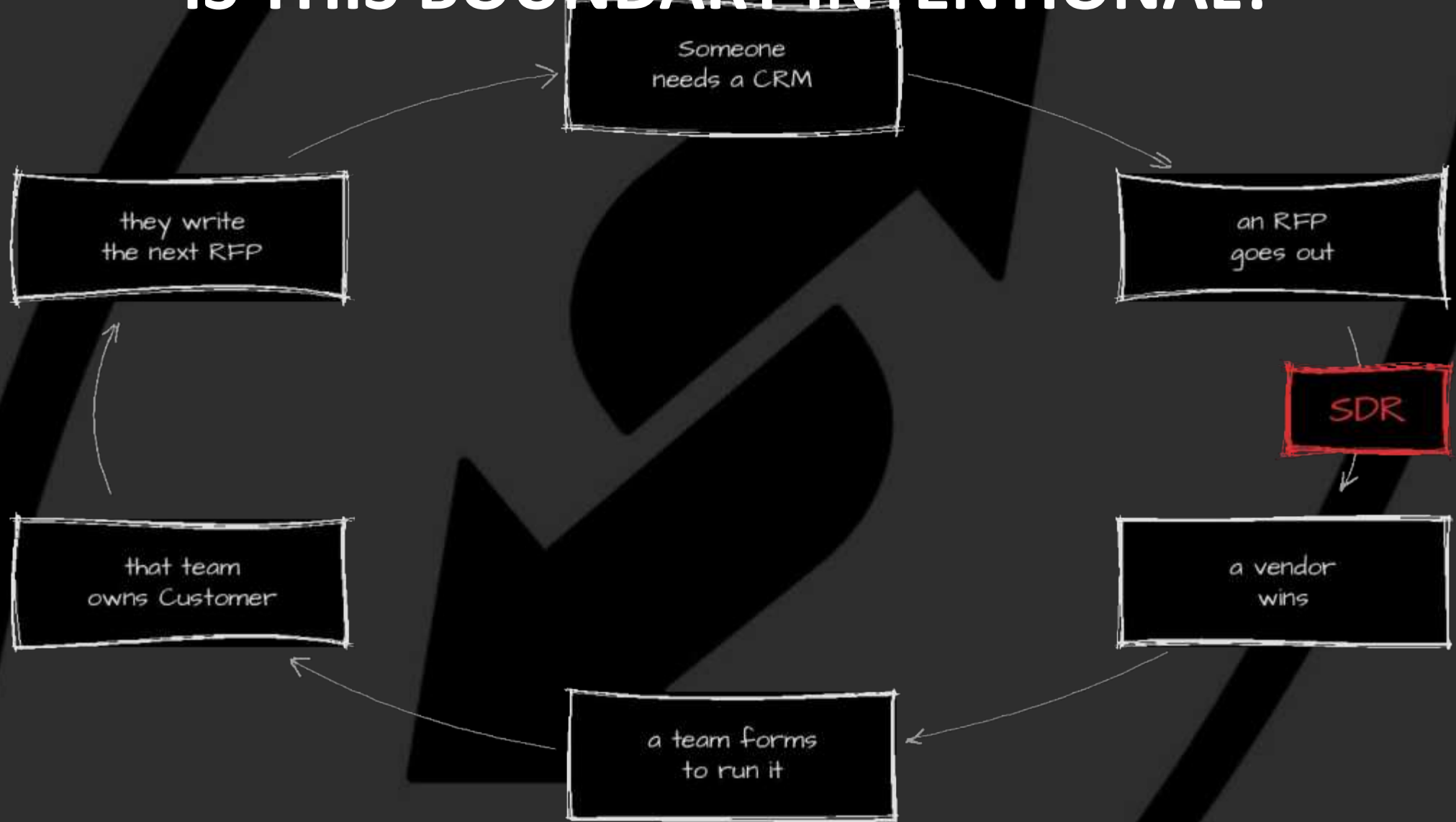
Preserve implementation reasoning.

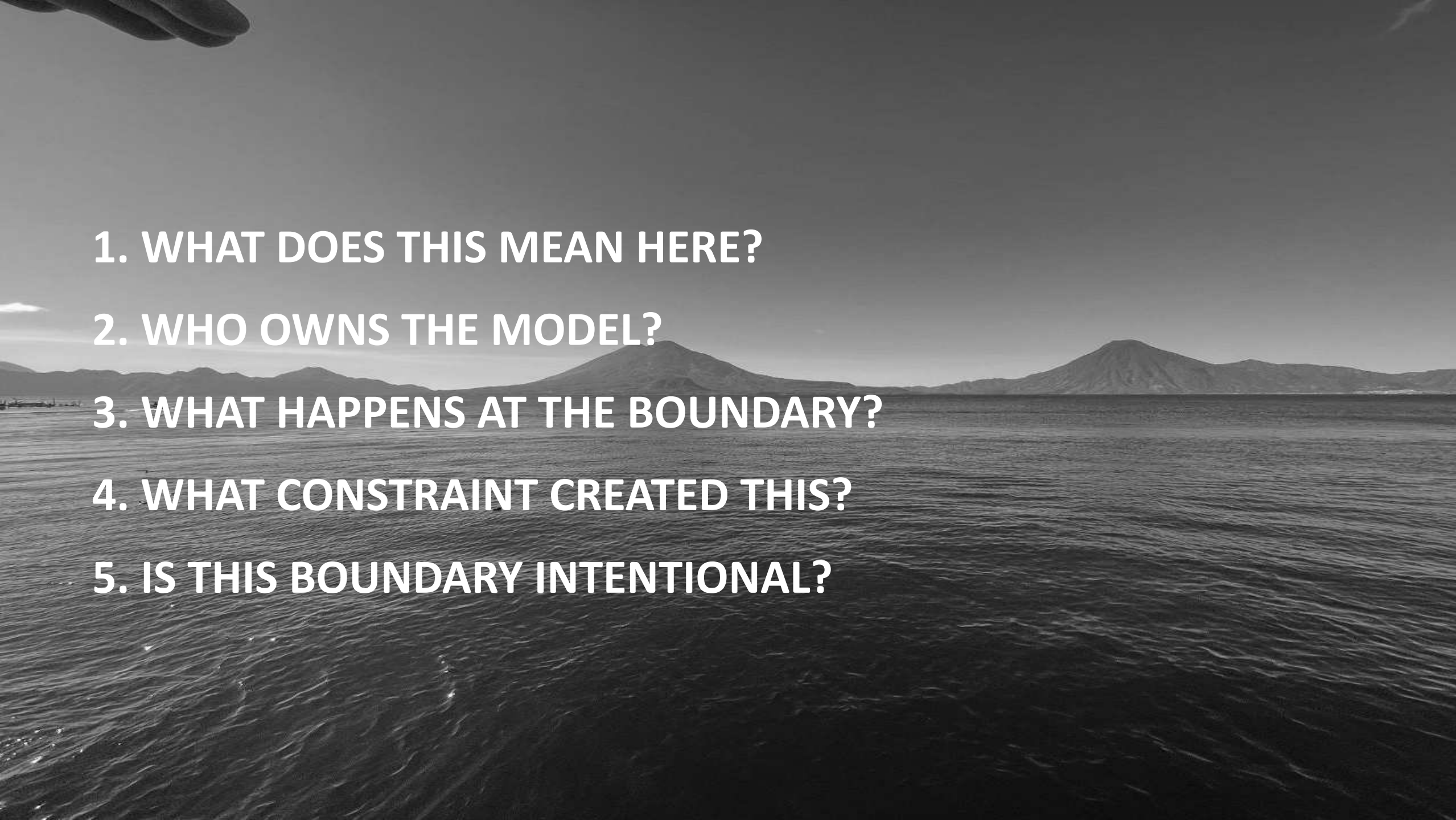
SDR

Why is this context downstream
of that one?

Preserve strategic reasoning.

IS THIS BOUNDARY INTENTIONAL?



- 
1. WHAT DOES THIS MEAN HERE?
 2. WHO OWNS THE MODEL?
 3. WHAT HAPPENS AT THE BOUNDARY?
 4. WHAT CONSTRAINT CREATED THIS?
 5. IS THIS BOUNDARY INTENTIONAL?

Same architecture, different eyes







A hand-drawn sketch of a terminal window. The title bar at the top is shaded with diagonal lines and contains the text "BANK CORE v2.1" on the left and "LOAN INQ" on the right. Below the title bar, there is a dashed horizontal line. Underneath the line, the following text is displayed: "LOAN NO 7731-0092", "PRINCIPAL 250,000.00", "RATE 6.25%", and "TERM 360". At the bottom of the window, the text "F3=Exit F5=Refresh F12=Cancel" is shown.

```
BANK CORE v2.1      LOAN INQ
-----
LOAN NO ..... 7731-0092
PRINCIPAL ..... 250,000.00
RATE ..... 6.25%
TERM ..... 360

F3=Exit F5=Refresh F12=Cancel
```

THEN

Understand the bank

Discover the model

Design the boundary

Choose technology

NOW

Understand the spaces between systems

Discover where language stops

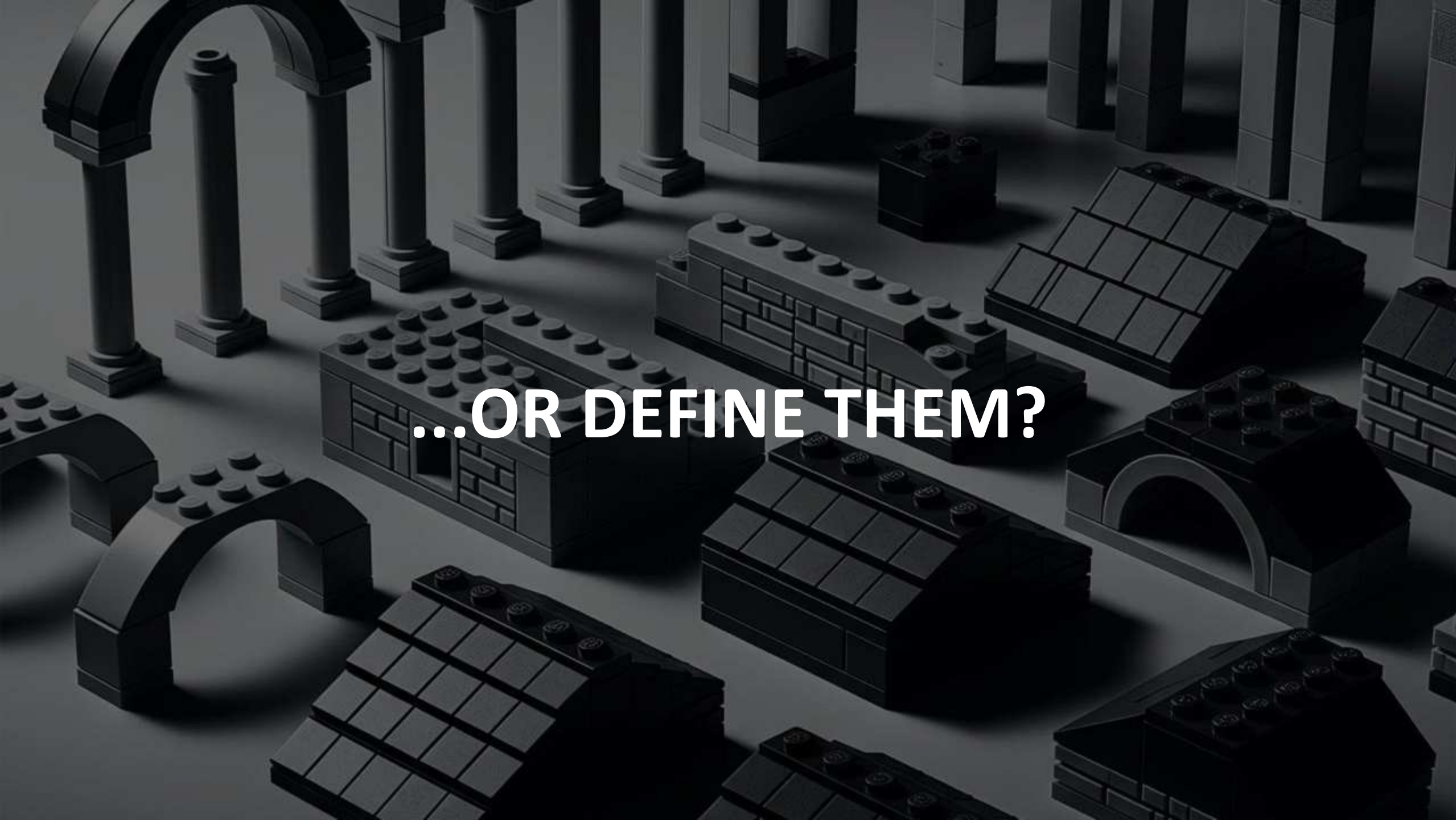
Negotiate ownership

Technology may already exist





**DO OUR PLATFORMS
IMPLEMENT OUR DOMAINS?**



...OR DEFINE THEM?

A hand-drawn illustration of a book cover. The cover is a solid blue rectangle with a white border. The title 'Domain-Driven Design' is written in white, sans-serif font, centered on the cover. The author's name 'Eric Evans' is written in a smaller white, sans-serif font at the bottom center. The spine of the book is on the left, indicated by a vertical white line and some light blue hatching. The entire book is centered on a black background.

Domain-
Driven
Design

Eric Evans

WHERE DOES MEANING LIVE?

DANKE!
THANK YOU!
MERCI!
GRAZIE!
GRACIAS!
DANK JE WEL!

.